

```
create database joinsdb;

use joinsdb;

CREATE TABLE Departments (
  DepartmentID INT PRIMARY KEY,
  DepartmentName VARCHAR(50)
);

INSERT INTO Departments (DepartmentID, DepartmentName)
VALUES
(101, 'HR'),
(102, 'IT'),
(103, 'Finance'),
(104, 'Marketing');

CREATE TABLE Employees (
  EmployeeID INT PRIMARY KEY,
  EmployeeName VARCHAR(50),
  DepartmentID INT
);

INSERT INTO Employees (EmployeeID, EmployeeName, DepartmentID)
VALUES
(1, 'Alice', 101),
(2, 'Bob', 102),
(3, 'Charlie', 105), -- No matching department
(4, 'David', NULL), -- No department assigned
(5, 'Eva', 101);


select* from Departments;

select * from employees;
```

```
select e.employeeid, e.employeename, e.departmentid,  
d.departmentid, d.departmentname  
from employees as e join department as d  
on e.employeeid =d.departmentid;
```

```
select e.employeeid, e.employeename, e.departmentid,  
d.departmentid, d.departmentname  
from department as d left join employees as e  
on e.departmentid=d.departmentid;
```

```
select e.employeeid, e.employeename, e.departmentid,  
d.departmentid, d.departmentname  
from department as d left join department as d  
on e.departmentid=d.departmentid;
```

-- 1. Primary Key (PK) =

-- A Primary Key is the column that uniquely identifies each row in a table

-- 2. Foreign Key (FK) =

-- A Foreign Key is a column in one table that points to the Primary Key of another table.

-- It is the actual link, the rope tying two tables together

```
use world;
```

```
select * from city;
```

```
select * from country;
```

```
select c.id , c.name, c.countrycode, city.code, city.name from city as c  
join country as city  
on c.countrycode=city.code;
```

```
-- cross join /cartesian join ==
```

```
select c.id , c.name, c.countrycode, city.code, city.name from city as c  
cross join country as city;
```

```
select c.id , c.name, c.countrycode, city.code, city.name from city as c  
join country as city;
```

```
select count(*) from city; -- 4079
```

```
select count(*) from country; -- 239
```

```
select count(*) from city as c  
join country as city;
```

```
select city.code, city.name from country as city;
```